Large Cardinals

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Preface

These notes are merely a comprehensive and filtered summary of references [3], [1] and [2], along with some of my own intuitions that may be useful to someone, and some additional comments on things I've found elsewhere.

Summary

CHAPTER 1

"Small" Large Cardinals

There are a lot of ways to motivate the study of Large Cardinals, and we will do it while progressively presenting them, but you can see one of them in my Cofinality notes. There I explain how we can come up naturally with the concept of a weakly inaccessible cardinal, and showed some properties of it. Before remembering some of them, let's see a toy model example of what we mean by a Large Cardinal Axiom.

As we will see, Large Cardinals are so large that their existence can't be proved in ZFC alone. But what if I told you we already have[1] some kind of large cardinal axiom in ZFC?

Imagine yourself as an ultrafinitist, you don't believe in such things as infinite cardinals, but this doesn't stop you from believing that Set Theory is important. In fact you use set theory a lot! But just as a branch of combinatorics, where everything is purely hereditarily finite.

Hence, what can we say about the Axiom of Infinity, other than it's pointless? Or even better, that it's utter bullshit! I mean, when does anyone used that? So you work in a weaker theory, which we will call FST (short for Finite Set Theory).

As we know, $R(\omega) = V_\omega \vDash \text{FST}$, and $\omega \notin V_\omega$, so FST can't prove there are such things as infinite cardinals. Even more, we have that the existence of such cardinal implies the consistency of FST! So we could say the Axiom of Infinity is some kind of Large Cardinal Axiom, but for ultrafinitist. Even though this was a nice analogy, you don't need to keep thinking like an ultrafinitist for now[2].

As a final remark, just to pin down, every time we state a consistency result, like $T \nvdash \varphi$, we're implicitly assuming $\text{Con}(T)$, or whatever we need to get rid of the trivial cases. Furthermore, Large Cardinal Axioms will be some sentence of the form $\exists\kappa(\Phi(\kappa))$, which we will denote by ΦC.

1.1 Inaccessible Cardinals

I will treat the following section as a direct continuation of my Cofinality notes. Therefore, any term or reference to any result not explicitly stated here can be consulted there.

[1] with a lot of poetic license

[2] fortunetaly, otherwise you wouldn't get past the first page

1.1.1 Fixed Point Lemma for Normal Functions

As we saw, a weakly inaccessible cardinal κ is just a uncountable limit regular cardinal. One of its properties is that it's an aleph fixed point, meaning $\aleph_\kappa = \kappa$. However, in my cofinality notes I proved that being an aleph fixed point is not too strong, since we can prove there are arbitrarily large aleph fixed points. As promised, I will show here a more general result.

Definition 1.1: Normal Function

A class-function $F : \mathrm{Ord} \to \mathrm{Ord}$ is called normal if it's both:
- monotone increasing, i.e., $\alpha < \beta \Rightarrow F(\alpha) < F(\beta)$;
- continuous in the order topology, i.e., if $\lambda > 0$ is a limit ordinal, then $F(\lambda) = \bigcup_{\alpha<\lambda} F(\alpha)$.

Then we can prove the following important Lemma

Lemma 1.1: Fixed Point Lemma for Normal Functions

The class of fixed points of any normal class-function F is unbounded in Ord.

Note then that, since $F(\alpha) = \aleph_\alpha$ is a normal function, such theorem implies there are arbitrarily large aleph fixed points. In fact, the idea of the proof is similar of the one for aleph fixed points.

Proof. Given an ordinal α, we want to let $\beta = \bigcup_{n\in\omega} F^n(\alpha)$, then, since F is increasing, $\beta \geq \alpha$. We now wish to use normality to conclude that

$$F(\beta) = \bigcup_{n\in\omega} F(F^n(\alpha)) = \bigcup_{n\in\omega} F^{n+1}(\alpha) = \beta$$

In fact, let's prove something stronger: If A is a non-empty set of ordinals, then

$$F\left(\bigcup_{\alpha\in A} \alpha\right) = \bigcup_{\alpha\in A} F(\alpha)$$

($\supseteq$) Since $\alpha \leq \bigcup_{\alpha\in A} \alpha$, and F is increasing, then $\bigcup_{\alpha\in A} F(\alpha) \leq F\big(\bigcup_{\alpha\in A} \alpha\big)$.

($\subseteq$) Let $\beta = \bigcup_{\alpha\in A} \alpha$, if β is a successor ordinal, then β is already in A, so the inclusion is trivial. Otherwise, β is a limit ordinal, and then by normality we have

$$F(\beta) = \bigcup_{\lambda<\beta} F(\lambda) \leq \bigcup_{\alpha\in A} F(\alpha)$$

Where the last inequality come from the fact that if $\lambda < \beta = \bigcup_{\alpha\in A} \alpha$, then $\lambda < \alpha$, for some $\alpha \in A$, and since F is increasing, $F(\lambda) < F(\alpha)$. ■

And, now we have an appropriate nomenclature, note we can prove an inaccessible κ is a F fixed point, for some other normal functions F:

For example, whenever $F(\alpha) < \kappa$, for any $\alpha < \kappa$.

Since F is increasing, $F(\alpha) \geq \alpha$, for any ordinal α. But also, since F is normal

$$F(\kappa) = \sup_{\alpha<\kappa} F(\alpha) \leq \sup_{\alpha<\kappa} \alpha = \kappa$$

so $F(\kappa) = \kappa$. Then, for example, κ would also be a beth fixed point, since we can let $F(\alpha) = \beth_\alpha$. Also, if κ is already a F fixed point, we can let G be an enumeration of the F-fixed points, and then κ would also be a G-fixed point. So for example, if you list all aleph fixed point, κ would be as far in the "line" of such fixed points as its own size, and you can keep going on such process.

1.1.2 Hausdorff's Theorem

Let $\mathrm{WI}(\kappa)$ be "κ is a weakly inaccessible cardinal", and WIC be the assertion that $\exists\kappa(\mathrm{WI}(\kappa))$. It's actually a non-trivial fact, which we will show at the end of the chapter, that

$$\mathrm{ZFC} \nvdash \mathrm{WIC} \tag{1}$$

but, for know, we can show this for (strongly) inaccessible cardinals. Let then $\mathrm{I}(\kappa)$ denote the statement that κ is an inaccessible cardinal and $\mathrm{IC} = \exists\kappa(\mathrm{I}(\kappa))$, as usual.

We will now show that ZFC can't prove inaccessible cardinals exists. This is done by proving $V_\kappa \vDash \mathrm{ZFC}$, when κ is an inaccessible cardinal. As usual, we first need to stablish a bound on the rank of sets constructed from V_κ. This is done by the next lemma.

Lemma 1.2

If κ is an inaccessible cardinal, then $|x| < \kappa$, for every $x \in V_\kappa$.

Proof. We will give two distintic proofs for this lemma:

- In the first, we will use a result in my Consistecy Results notes, which says $|V_\gamma| \le \beth_\gamma$. Now, given κ an inaccessible, and $x \in V_\kappa$, then $x \in V_\gamma$ for some $\gamma < \kappa$, but since V_γ is a transitive set, then we also have $x \subseteq V_\gamma$, so
$$|x| \le |V_\gamma| \le \beth_\gamma < \beth_\kappa = \kappa$$
where in the next equality we used that κ is a beth fixed point, as we saw in the end of the previous subsection;
- For the second, we will use transfinite induction on γ to show $|V_\gamma| < \kappa$. For $\gamma = 0$ it's trivial. Now, assume it's true for γ, then, since κ is a strong limit,
$$|R_{\gamma+1}| = |\mathcal{P}(V_\gamma)| = 2^{|V_\gamma|} < \kappa.$$
Now, if λ is a non-zero limit ordinal, and for all $\gamma < \lambda$, $|V_\gamma| < \kappa$, then, since κ is regular,
$$|V_\lambda| = \left|\bigcup_{\gamma<\lambda} V_\gamma\right| \le \sum_{\gamma<\lambda} |V_\gamma| < \kappa.$$

■

We can now prove our main result, due to Hausdorff.

Theorem 1.1: (Hausdorff)

If κ is inaccessible, then $V_\kappa \vDash \mathrm{ZFC}$.

Proof. First of all, I already have a complete proof in my Consistency Results notes that $V_\gamma \vDash \mathrm{ZC}$, whenever $\gamma > \omega$ is a limit ordinal. Then it's sufficient to show V_κ satisfies replacement.

Now, as usual, we will prove the second order version of replacement, then let $F : V_\kappa \to V_\kappa$ and $x \in V_\kappa$, we claim $F[x] \in V_\kappa$. Since $F[x] = \{F(y) : y \in x\}$, let's estimate $\mathrm{rank}(F(y))$. Since V_κ is transitive, and $x \in V_\kappa$, then $y \in V_\kappa$, and so $F(y)$, therefore $\mathrm{rank}(F(y)) < \kappa$. Now, let

$$C = \{\mathrm{rank}(F(y)) + 1 : y \in x\} \subseteq \kappa$$

Since $|C| \leq |x| < \kappa$ (by the previous lemma), then C is a subset of κ of size $< \kappa$, then, since κ is regular, C is bounded in it by some α, and then $F[x] \in V_{\alpha+1} \subseteq V_\kappa$, as desired. ■

Note that 1 is now a direct corollary of Hausdorff's Theorem and Gödel's Incompleteness. However, we can also give a Gödel free proof of it:

Corollary 1.1

$$\mathrm{ZFC} \nvdash \mathrm{IC}.$$

Proof. If ZFC proved it, we could define the least inaccessible cardinal κ_0, and then, by Hausdorff's Theorem, $V_{\kappa_0} \vDash \mathrm{ZFC}$, in particular V_{κ_0} also satisfies that there is an inaccessible cardinal λ, but then $\lambda \in V_{\kappa_0}$, so $\lambda < \kappa_0$, contradicting the minimality of κ_0. ■

To be fair, the previous proof is not entirely correct, since we didn't justified why λ being inaccessible *in* V_{κ_0} implies that it's in fact inaccessible. In order to justify that, we need to show that the possible counterexamples for λ being a cardinal, regular and strong limit, would also be "seen" by V_{κ_0}:

Lemma 1.3

Working in $\mathrm{ZFC} + \mathrm{I}(\kappa)$, for any $\lambda < \kappa$ we can prove

$$\mathrm{I}(\lambda) \iff V_\kappa \vDash \mathrm{I}(\lambda)\,.$$

Proof. This can be rather tedious if you don't know nothing about absoluteness[3], but the idea is that $\mathrm{I}(\lambda)$ is a Π_1 formula, implying it's downward absolute under transitive models as V_κ.

For the converse, assume for example that λ is not a cardinal, then we can find $f : \alpha \to \lambda$ surjective, with $\alpha < \lambda$. Since $\lambda \in V_\kappa$, then $\lambda \in V_\beta$ for some $\beta < \kappa$, but then, by transitivity, $\alpha \in V_\beta$ too and, therefore, $f \in V_{\beta+3} \subseteq V_\kappa$, so λ wouldn't be a cardinal in V_κ too.

The proof is similar for the other possibilities, we just need to show our witness of failure will also be in V_κ, whose proof idea is similar. ■

1.2 Worldly Cardinals

As we saw, $\mathrm{I}(\kappa)$ imply $V_\kappa \vDash \mathrm{ZFC}$. Can we prove the converse, turning Hausdorff's theorem into a characterization? For the sake of investigation, let's then define

[3]you can learn more about absoluteness in my Consistency Results notes

Definition 1.2: Worldly Cardinal

A Worldly Cardinal κ is a cardinal such that $V_\kappa \vDash \mathrm{ZFC}$.

We denote by $\mathrm{W}(\kappa)$ the statement that κ is a worldly cardinal. As a warmup to see how we work with worldly cardinals, let's prove the following two lemmas.

Lemma 1.4

If an ordinal κ is such that $V_\kappa \vDash \mathrm{ZFC}$, then κ is worldly.

What the previous lemma says is that if the α-th level of the Von Neumann Hierarchy is a model of ZFC, then α itself is automatically a cardinal.

Proof. [4] If it's not, we would have a bijection $f : \lambda \to \kappa$, for some $\lambda < \kappa$. But then consider the following induced well-ordering in λ of order type κ:

$$W = \{(\alpha, \beta) : f(\alpha) < f(\beta)\}$$

Let's show κ is also a limit cardinal. In fact, if it's not, $\kappa = \mu + 1$ would be such that, if $\alpha \in V_{\mu+1}$, then $\mathrm{rank}(\alpha) \leq \mu$, i.e., μ would be a largest ordinal, contradicting that $V_\kappa \vDash \mathrm{ZFC}$.

Now, since $\lambda \in V_\kappa$, then $\lambda \in V_\beta$, for some $\beta < \kappa$, then $W \in V_{\beta+3}$ and, therefore, $(\lambda, W) \in V_{\beta+5} \subseteq V_\kappa$. Since (λ, W) is a well-ordering, and $V_\kappa \vDash \mathrm{ZFC}$, it's in bijection with a unique ordinal, but such ordinal is κ itself! So $\kappa \in V_\kappa$, a contradiction. ■

And what can we say about the least worldly cardinal? Surprisingly, we can prove the following.

Lemma 1.5

The least worldly cardinal κ has countable cofinality.

Proof. Assume to reach a contradiction that $\mathrm{cf}(\kappa) > \aleph_0$, and let $\mathrm{ZFC} = \{\varphi_n\}_{n\in\omega}$, $\Phi_n = \{\varphi_k\}_{k\leq n}$. Since $V_\kappa \vDash \mathrm{ZFC}$, for each $n \in \omega$ we get, by Reflection Theorem[5], a club in κ:

$$C_n = \{\alpha < \kappa : V_\alpha \preceq_{\Phi_n} V_\kappa\}$$

It's a well known theorem that the Club Filter in κ is $\mathrm{cf}(\kappa)$-complete[6], so

$$C = \bigcap_{n\in\omega} C_n$$

is also a club. Now, let $\alpha \in C$, then $V_\alpha \preceq_{\Phi_n} V_\kappa$, for every $n \in \omega$, which implies $V_\alpha \vDash \mathrm{ZFC}$, contradicting that κ is the least worldly ordinal. So $\mathrm{cf}(\kappa) = \aleph_0$. ■

You can interpret the previous lemma as saying that, at least, the first worldly cardinal is "small", which naturally lead you to the question of what are the possible cofinalities of worldly cardinals. In fact, the next theorem will have everything you need to answer this question.

[4] it's worth noticing that, as before, we use some absoluteness results to go from V to V_κ

[5] you can see a full proof of it in my Consistency Results notes, in the Reflection Theorem section

[6] I have a proof for κ regular in my Cofinality notes, but it's clear how to generalize it to any cardinal κ instead

Theorem 1.2

If κ is an inaccessible cardinal, then $\{\lambda < \kappa : V_\lambda \preceq V_\kappa\}$ is unbounded in κ.

Sketch: Given $\alpha < \kappa$, the idea to find such $V_\lambda \preceq V_\kappa$, for $\alpha < \lambda < \kappa$, is to use Tarski Vaught's Test[a] (TVT). In order to do that, we start with V_α and, for each $\varphi(x, \overline{a})$, with $\overline{a} \in V_\alpha^{<\omega}$, if $V_\kappa \vDash \varphi(b, \overline{a})$, for some $b \in V_\kappa$, we find the level β where $b \in V_\beta$, and then we consider the limit process of taking all such β's (it's not hard to see how regularity of κ helps us here).

We then just do a ω-iteration of such process to construct our elementary substructure V_λ that is closed under such witnessing required by TVT.

Proof. Given $\alpha < \kappa$, let $\alpha_0 = \alpha + 1$. By recursion, assume $\alpha_n < \kappa$ is already defined. We know by lemma 1.2 that $|V_{\alpha_n}| < \kappa$, so $|V_{\alpha_n}^{<\omega}| < \kappa$ too. Furthermore, since there are only countably many formulas in the language of Set Theory, we have

$$\left|\mathrm{Fml} \times V_{\alpha_n}^{<\omega}\right| < \kappa \tag{2}$$

It's clear that $\mathrm{Fml} \times V_{\alpha_n}^{<\omega}$ encodes our formulas $\varphi(x, \overline{a})$, where $\overline{a} \in V_{\alpha_n}$. Let's now define our rank function r in such set of formulas that gives us the level of our witness.

Given $\varphi(x, \overline{a})$, if $V_\kappa \vDash \varphi(b, \overline{a})$, for some $b \in V_\kappa$, then $b \in V_\gamma$, for some $\gamma < \kappa$, in this case set $r(\varphi, \overline{a}) = \gamma$. Otherwise, just let $r(\varphi, \overline{a}) = 0$.

Now, since the domain of r satisfies (2), it's a set of $< \kappa$ ordinals $< \kappa$. By regularity of κ, it's bounded by some $\beta < \kappa$, and then we can deefine

$$\alpha_{n+1} := \max\{\beta, \alpha_n + 1\}$$

Then, we just need to let

$$\lambda := \bigcup_{n\in\omega} \alpha_n < \kappa$$

where the inequality also follows from regularity. Then, clearly, if $V_\kappa \vDash \varphi(b, \overline{a})$, for some $b \in V_\kappa$ and $\overline{a} \in V_\lambda^{<\omega}$, then $\overline{a} \in V_{\alpha_n}^{<\omega}$, for some $n \in \omega$, which, by definition, implies that there is some $c \in V_{\alpha_{n+1}}^{<\omega} \subseteq V_\lambda^{<\omega}$ such that $V_\lambda \vDash \varphi(c, \overline{a})$. So, by TVT, $V_\lambda \preceq V_\kappa$. ∎

Let's see what are the consequences of such theorem. First, note that, if κ is worldly, and $V_\lambda \preceq V_\kappa$, then λ is also worldly. So the previous theorem actually implies that the set of worldly cardinals bellow an inaccessible is unbounded in it! Which means, for example, that there are as many worldly cardinals bellow κ as we can, i.e., exactly κ many. Maybe more impressive, if we look at the proof of the theorem, we can conclude the following as a scholium.

Corollary 1.2

The set of worldly cardinals of cofinality $\aleph_0$ bellow an inaccessible κ is unbounded in it.

Such corollary have two important consequences hidden inside it. The first is the answer for the question we posed at the beginning of the section. If κ is the least inaccessible cardinal, by the previous Theorem we get κ-many smaller worldly, and then, even though $\mathrm{I}(\kappa) \Rightarrow \mathrm{W}(\kappa)$, the converse

[a] as always, it's in my Consistency Results notes

is unattainable. However, this doesn't mean we have no hope to turn Hausdorff's Theorem into a characterization! In fact, the next lemma shows we can.

Lemma 1.6: (Zermelo and Shepherdson)

κ is an inaccesible cardinal if, and only if, $V_\kappa \vDash \mathrm{ZFC}_2$.

where ZFC_2 denotes the second order version of ZFC, with the replacement schema replaced by its single second order analogue.

Proof. ($\Rightarrow$) This is exactly the proof of Hausdorff's Theorem;

($\Leftarrow$) To see κ is regular, assume it's not, then there would be an $F : \alpha \to \kappa$ with image unbounded in κ, but since $F \subseteq V_\kappa$ and we have second order replacement, then $F[\alpha] \in V_\kappa$. Hence $\sup(F[\alpha]) = \kappa \in V_\kappa$, a contradiction.

Then it remains to show it's a strong limit. If there was a $\lambda < \kappa$, with $2^\lambda \geq \kappa$, then we would have a surjection $F : \mathcal{P}(\lambda) \to \kappa$, but $\mathcal{P}(\lambda) \in V_\kappa$, meaning $F \subseteq V_\kappa$ too and, again, by second order replacement $F[\mathcal{P}(\lambda)] = \kappa \in V_\kappa$, a contradiction. ■

And the second is that there is no chance to prove that every inaccessible cardinal will have a smaller regular worldly cardinal, as the next corollary, jointly with the relative consistency of GCH[7], shows.

Corollary 1.3

Under GCH + IC, there are inaccessible cardinals with no regular worldly cardinals below it.

Proof. Under GCH we have that limit and strong limit are equivalent, and then, if there was any regular worldly bellow the least inaccessible, it couldn't be a limit cardinal, since this would imply it's also inaccessible. However, as we saw in the proof of lemma 1.4, worldly cardinals are always limit, and then we don't have any worldly cardinal bellow the least inaccessible. ■

As for now, what we saw is that there are a lot of "small" worldly cardinals bellow an inaccessible, and that there is no hope to find "very large" ones, like regulars. A natural question is then for which other cofinalities we can attain analogous results?

In fact, note that the quirk of having countable cofinality is just a consequence of we choosing a countable increasing sequence (α_n) of ordinals, why couldn't we just do the same for other possible lengths of our sequence? It turns out we can, and for almost all possible lengths!

Corollary 1.4

If κ is inaccessible, and $\mu < \kappa$ is regular, then the set

$$W^\mu := \{\lambda < \kappa : \mathrm{W}(\lambda) \text{ and } \mathrm{cf}(\lambda) = \mu\}$$

is unbounded in κ.

So, even thought we can find many worldly cardinals with arbitrarily large cofinalities bellow an inaccessible κ, we can't guarantee there would be even a single regular.

[7] for a full proof that GCH is consistent with ZFC, look at The Constructible Universe L chapter in my Consistency Results notes

Proof. by theorem 1.2 we can let

$$(\lambda_\alpha)_{\alpha<\kappa} = \{\lambda \in \kappa : V_\lambda \preceq V_\kappa\}$$

then, $V_{\lambda_\alpha}, V_{\lambda_\beta} \preceq V_\kappa$, for $\alpha, \beta \in \kappa$ and, if $\alpha < \beta$, then $V_{\lambda_\alpha} \preceq V_{\lambda_\beta}$. So, if $X \subseteq \kappa$ is an arbitrary subset, we have that $\{V_{\lambda_\alpha} : \alpha \in X\}$ is an elementary chain.

Now, consider $X = \mu$ regular, then $\bigcup_{\alpha<\mu} V_{\lambda_\alpha} = V_\lambda$, where $\lambda = \bigcup_{\alpha<\mu} \lambda_\alpha$ and, by Tarski Chain Lemma[8], we get $V_{\lambda_\alpha} \preceq V_\lambda$ for any $\alpha < \mu$, so $V_\lambda \vDash$ ZFC. But then, by definition of λ, we automatically have that $\mathrm{cf}(\lambda) = \mu$, as desired. ■

1.3 Weakly Inaccessible Cardinals

As I promised in section 1.1.2, here is a proof that ZFC can't prove there is a weakly inaccessible cardinal.

Theorem 1.3

ZFC $\nvDash$ WIC.

Proof. Here we will need some of the same results we used throughout the last section, namely the constructible universe L. If there was a weakly inaccessible κ, we could consider L_κ. We already know for uncountable regular μ that $L_\mu \vDash$ ZFC − PowerSet. Now, by a similar argument to V_κ, given $x \in L_\kappa$, we know $x \in L_\lambda$, for some $\lambda < \kappa$, but then $\mathcal{P}(x) \cap L_\kappa \subseteq L_{\lambda+1} \subseteq L_\kappa$, and then $L_\kappa \vDash$ ZFC so we can prove ZFC is consistent, a contradiction. ■

Until now, we saw IC implies both WIC and WC, and that IC is not equivalent to WC, then what can we say about WIC? Here we run in some problems: in the proof that WC $\nRightarrow$ IC, we showed that, if κ was inaccessible, we could always find a smaller worldly that was not inaccessible. However, what if we tried to do the same for WIC? Let κ be an inaccessible, and suppose we make some construction to find a weakly $\mu < \kappa$ that is not inaccessible, then we could repeat the exactly same construction in L! And then μ would also be an inaccessible, a contradiction.

In other words, we can't search for weakly inaccessibles inside the model, we need to look outside, i.e., using Forcing! As I kept saying through this first chapters, almost all consistency tool you don't know you can look in my Consistency Results note that you will probably find it there, this is no different for Forcing.

So let κ be an inaccessible cardinal. Since it has uncountable cofinality, we can use Hechler's Forcing to force $\mathfrak{c} = \kappa^+$, by ccc κ is still a limit and regular, but now, since $\aleph_0 < \kappa$, and

$$2^{\aleph_0} = \mathfrak{c} = \kappa^+ > \kappa$$

then κ is no longer strongly inaccessible! As desired.

So now it remains to pin the relationship between WIC and WC. Well, under WIC, as we saw, the least worldly has countable cofinality, and therefore can't be weakly inaccessible.

[8] this is a classical lemma whose proof is just an adaptation of the proof of TVT

CHAPTER 2

Larger Large Cardinals

2.1 The Measure Problem

2.1.1 Set Theoretic Version

It's a classical problem to seek for a non-constant, translational invariant, σ-additive probabilistic measure $\mu : \mathcal{P}(\mathbb{R}) \to [0,1]$ in the reals $\mathbb{R}$. And, as we know, Vitali showed such task is impossible.

The construction is rather simple for a modern set theorist, consider the real numbers modulo a rational, $\mathbb{R}/\mathbb{Q}$, and let V be a transversal in $[0,1]$. We will show

$$[0,1] \subseteq \bigcup_{q \in [-1,1]\cap\mathbb{Q}} V + q \subseteq [-1,2].$$

Since $V \subseteq [0,1]$, the second inclusion is clear. For the first let $r \in [0,1]$, then by definition of V we can find $v \in [r] \cap V$, so $r - v \in [-1,1] \cap \mathbb{Q}$, and then $r = v + (r - v) \in V + (r - v)$. So

$$0 < \mu([0,1]) \leq \sum_n \mu(V) \leq \mu([-1,2]) < 1$$

which is impossible for any value $\mu(V)$ could assume.

From here, there are two possible paths. Measure theory lead the one weakening the domain of μ, i.e., considering the measure only for the sets in a σ-algebra of $\mathbb{R}$. Set theory lead the other where the geometrical condition of translational invariance is weakened.

Banach showed that, without translational invariance, we could find trivial measures μ_r on any set S, with $r \in S$, as follows:

$$\mu_r(A) = \begin{cases} 1, & \text{if } r \in A; \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

So he imposed that, instead of translational invariance, we had a non-triviality condition, i.e., that $\mu(\{x\}) = 0$, $\forall x \in S$ which, with σ-additivity, guarantees us that all finite sets have measure 0 too.

So, for the sake of our sanity, let's just pin down that, throughout this section, a function $\mu : \mathcal{P}(S) \to [0,1]$ is a measure on/over S if $\mu(\emptyset) = 0$, $\mu(S) = 1$, it's non-trivial and it's σ-additive.

A measure on S depends only on $|S|$, so for now on we will consider measures solely on cardinals. We can then state the set-theoretic version of the Measure Problem, as proposed by Banach:

Set Theoretic Measure Problem (Banach)

Can we find a measure on $\mathfrak{c}$. Or rather at least on some infinite κ?

In order to keep investigating measures on infinite cardinals, consider the following generalization of the notion of σ-additiveness, which guarantees us that, as for the finite sets, any set of cardinality $\lambda < \kappa$ has also measure zero.

Definition 2.1: κ-additive measure

We say a measure μ is κ-additive if, for any $\lambda < \kappa$ and pairwise disjoint sets $\{X_\alpha\}_{\alpha<\lambda}$,

$$\mu\left(\bigcup_{\alpha<\lambda} X_\alpha\right) = \sum_{\alpha<\lambda} \mu(X_\alpha) := \sup\left\{\sum_{\alpha\in E} \mu(X_\alpha) : E \in [\lambda]^{<\omega}\right\}.$$

With such nomenclature, σ-additive is just $\aleph_1$-additive.

In fact, if μ is a κ-additive measure on a set S, then any subset of size $< \kappa$ has measure 0, as desired. The next lemma explains why we care about κ-additiveness, it says that the solution for the Set Theoretic Measure Problem is related to the existence of a κ-additive measure.

Lemma 2.1

Let κ be the least cardinal with a measure in it, then every measure over κ is κ-additive.

Proof. Assume by a contradiction that there are pairwise disjoint $\{X_\alpha\}_{\alpha<\lambda}$ with

$$\mu\left(\bigcup_{\alpha<\lambda} X_\alpha\right) \neq \sum_{\alpha<\lambda} \mu(X_\alpha). \tag{4}$$

As I said, $\mu(X_\alpha) \neq 0$ for at most countably many α's, so assume WLOG (by removing and reindexing them if necessary) that $\mu(X_\alpha) = 0$ for any $\alpha < \lambda$. But, by eq. (4), we have

$$\mu\left(\bigcup_{\alpha<\lambda} X_\alpha\right) = r > 0.$$

So the following $\widetilde{\mu} : \mathcal{P}(\lambda) \to [0, 1]$ is a measure over λ:

$$\widetilde{\mu}(Y) := \frac{1}{r} \cdot \mu\left(\bigcup_{\alpha\in A} X_\alpha\right)$$

contradicting the minimality of κ. ■

So, now that we saw the importance of κ-additive measures, we can then define:

Definition 2.2: Real-Valued Measurable

A cardinal $\kappa > \omega$ is real-valued measurable if there is a nontrivial κ-additive measure on κ.

Since a solution for the Set Theoretic Measure Problem implies the existence of a real-valued measurable cardinal, we can now ask if there is any real-valued measurable cardinal.

As expected of a Large Cardinal Notes, the answer for this problem is related to the existence of some large cardinals, and therefore is independent of ZFC. To see this, note that, if κ is real-valued measurable, by κ-additivity we have that κ is regular. In fact, Banach established that, under GCH, every real-valued measurable cardinal is weakly inaccessible. But we can do even better! Ulam showed in his doctoral dissertation that we can remove GCH from Banach's result, as we will see later.

2.1.2 Measures with Atoms

Before proving the previous results, let's remark an important dichotomy between measures:

Definition 2.3: Atoms

Let κ be real-valued measurable. A subset $A \subseteq \kappa$ with $\mu(A) > 0$ is called an atom if for every $B \subseteq A$, either $\mu(B) = 0$ or $\mu(B) = \mu(A)$.
We say a measure μ over κ is atomless if there is no atom $A \subseteq \kappa$.

What is so important about being or not atomless? Note that, if μ has an atom $A \subseteq \kappa$, then

$$\widetilde{\mu}(X) = \frac{\mu(A \cap X)}{\mu(A)}$$

is a 2-valued measure. We clearly have a correspondence, since every A with positive measure is an atom in a 2-valued measure. So, in some sense, atoms gives us some sort of generalization of Banach's trivial measure (3). We will see, however, that such measures are far from being trivial, and, maybe unexpectedly, will be way more fruitful to us than atomless measures!

As a matter of comparison, you could ask yourself if atomless measure can be as trivial as 2-measures; for example, if it could be a 3-measure. Fortunately, it's a theorem of Sierpiński that, if μ is atomless, given $X \subseteq S$ and $0 \leq \alpha \leq \mu(X)$, there is some set $Y \subseteq X$ such that $\mu(Y) = \alpha$. In other words, μ attains all possible values it could possibly reach.

So, the reason why atoms are such a big deal is that we can make a correspondence between κ-complete free ultrafilters on κ and κ-additive non-trivial measures with atoms on κ, as follows:

- Given an ultrafilter U, let $\mu(X) = 1$ iff $X \in U$;
- Given a measure μ, let $X \in U$ iff $\mu(X) = 1$.

Which gives rise to the following Large Cardinal Axiom.

Definition 2.4: Measurable Cardinal

A cardinal κ is measurable if there is a 2-valued κ-additive measure on it.

As noted earlier, κ-additivity implies regularity of κ. In fact, with atoms we can prove even more!

Theorem 2.1

Every measurable cardinal is inaccessible.

Proof. Let κ be a measurable cardinal and U be its κ-complete free ultrafilter.

Assume by a contradiction that there is some $\lambda < \kappa$, with $2^\lambda \geq \kappa$. Then we can find an injection $F : \kappa \to 2^\lambda$ such that $F_\alpha : \lambda \to 2$. So, given $\xi \in \lambda$, we have

$$\kappa = \{\alpha \in \kappa : F_\alpha(\xi) = 0\} \sqcup \{\alpha \in \kappa : F_\alpha(\xi) = 0\}.$$

But, since U is an ultrafilter, either one of these sets are in U. Let b_ξ be such that

$$A_\xi = \{\alpha \in \kappa : F_\alpha(\xi) = b_\xi\} \in U.$$

Then, by κ-completeness of U, we have that

$$A = \bigcap_{\xi<\lambda} A_\xi \in U.$$

However, if $\alpha \in A$, then $F_\alpha(\xi) = b_\xi$ for every $\xi \in \lambda$, i.e., F_α is a completely determined function, so $|A| \leq 1$, contradicting that U is free and $A \in U$. ■

This guarantees us that measurability is a Large Cardinal Axiom. Whether this is equivalent or not to inaccessibility was an open problem for quite some time. As expected (otherwise we weren't spent a whole section talking about measurables) they're not equivalent, which we will prove later after introducing some other large cardinals axioms.

2.1.3 Real-Valued Measures

Finally, as promised, let's return to the study of real-valued measures and see how the existence of atomless measures gives us some weird results about the continuum.

The previous subsection shows us that there is no hope to find[1] a 2-valued measure on $\mathfrak{c}$, but what about an atomless one? As we spoiled earlier, real-valued measurability implies weakly inaccessibility, which is now clear for the case with atoms. In order to prove this for real-valued measures, we can't use the set of subsets with measure 1, since it won't be a filter anymore.

So, to do what we want, let's analyze it's dual notion! I.e., the ideal of null subsets of κ:

$$I_\mu := \{X \subseteq \kappa : \mu(X) = 0\}.$$

It's clear that, if μ is a λ-additive measure, then I_μ is a free λ-complete ideal. Morever, it's what we call a σ-saturated ideal, i.e., every family $\mathcal{F}$ of pairwise disjoint subsets of κ that aren't in I_μ is at most countable. To see this, note that

$$\mathcal{F} = \bigcup_{n\geq 1} \{X \in \mathcal{F} : \mu(X) \geq 1/n\}$$

where each term in the RHS is at most finite.

This may seem like a not really powerful remark, but the fact that the ideal of null sets is σ-saturated is actually really strong! It's the reason why a lot of the following beautiful results work and also why we can relate the Set Theoretic Measure Problem with measures on $\mathbb{R}$. In fact, note that σ-saturation follows solely by the fact that we decided to measure sets with reals[2] in $[0, 1]$.

[1]at least in ZFC

[2]that's why it's called a real-valued measure, after all. despite κ-additivity, such generalization is not a completely set-theoretic one, since it relies on measuring as "humans" naturally would

About λ-additivity implying λ-completeness, we can also prove the converse result:

Lemma 2.2

Let μ be a measure on κ, if I_μ is λ-complete, then μ is λ-additive.

Proof. Given $\gamma < \lambda$, let $\{X_\alpha\}_{\alpha<\gamma}$ be a family of pairwise disjoint subsets of κ, since our ideal I_μ is σ-saturated, we know at most countably many of those X_α's have positive measure, let's say Y_n, for $n \in \omega$. But then, by κ-completeness of I_μ and σ-additivity of μ:

$$\mu\left(\bigcup_{\alpha<\gamma} X_\alpha\right) = \mu\left(\bigcup_{n\in\omega} Y_n\right) = \sum_{n\in\omega} \mu(Y_n) = \sum_{\alpha<\gamma} \mu(X_\alpha)$$

as desired. So μ is κ-additive. ■

Let's now prove our first consequence about the existence of atomless measures.

Theorem 2.2

If there is an atomless measure μ, then there is also one on some $\kappa \leq \mathfrak{c}$.

Sketch: Let's first sketch this proof, because it's really nice. It makes clear how different atomless measures are and how "strong" the σ-saturation property is.

Let μ be such measure on S. The fact that it's atomless gives us the following amazing property that resembles disconnectedness:

$$\text{If } X \subseteq S, \text{ then we can write } X = Y \sqcup Z, \text{ where } \mu(Y), \mu(Z) > 0$$

I.e., we can partition any set in subsets of positive measure. Otherwise X would be an atom.

Now, starting with S, we can partition it in subsets of positive measure, and then partition each of these subsets in other subsets of positive measure. If we keep going on such process, we end up with an infinite tree, where in the limit level we take all possible intersections of transversal with positive measure.

There is actually more than one way to construct such tree, we will explicit just one construction, thought. Let's now analyze how such tree looks like.

Each level of our tree consists of pairwise disjoint sets of positive measure, then, by σ-saturation, each level has countably many sets. What about the size γ of our tree? Let

$$b = \{X_\alpha : \alpha < \lambda\}, \text{ with } \lambda < \gamma \text{ and } X_\alpha \text{ a set in the } \alpha\text{-th level}$$

be branch of it. By definition, $\mu(X_{\alpha+1}) < \mu(X_\alpha)$, and so $Y_\alpha = X_{\alpha+1} \setminus X_\alpha$ form a family of λ-many pairwise disjoint sets of positive measure. Again, by σ-saturation, we have that the size of each branch is at most countable, and so the size γ of our tree is at most $\aleph_1$.

Now, look at the limit point of each branch, they all have measure 0, otherwise it wouldn't be the limit point, since we could keep going on our process. They also form a partition of S and, since it's a tree whose branches has countable height, and each level has countably many elements, there are at most $\mathfrak{c}$ of these limit points.

So we've constructed $\leq \mathfrak{c}$-many null sets, whose union is S, i.e., our measure isn't $\mathfrak{c}^+$-additive! Then it just remains to induce a measure on a smaller cardinal.

Proof. For this first construction we will make some kind of binary tree T. Start with S and, if the α-th level is already defined, given some set X in it, its immediate successors are the sets Y and Z of positive measure that partition X. And when α is a limit ordinal, we get all possible intersections $\bigcap_{\beta<\alpha} X_\beta$ with positive measure, where X_β is in the β-th level.

If it wasn't clear from the sketch, each branch has countable length in T, whose height is at most $\aleph_1$, so we can associate it branch with a countable subset of T, i.e., the amount κ of branches is st

$$\kappa \leq \left|[T]^{\aleph_0}\right| \leq \aleph_1^{\aleph_0} \leq \left(2^{\aleph_0}\right)^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = \mathfrak{c}$$

Let then $\{b_\alpha : \alpha < \kappa\}$ be an enumeration of all branches $b_\alpha = \{X_\beta : \beta < \gamma\}$ whose limit point

$$Z_\alpha := \bigcap_{\beta<\gamma} X_\beta$$

is nonempty. Then $\{Z_\alpha : \alpha < \kappa\}$ partition S in κ sets of measure 0.

We then induce a measure ν on κ as follows: Let $f : S \to \kappa$ be defined as

$$f(x) = \alpha \iff x \in Z_\alpha$$

and let

$$\nu(X) := \mu(f^{-1}(X)).$$

It's not hard to verify that ν is in fact a measure on κ. To see it can't have an atom, assume it has, then κ is measurable and, by theorem 2.1, it would be inaccessible. However, you can't have an inaccessible bellow the continuum, otherwise, since $\aleph_0 < \kappa \leq 2^{\aleph_0}$, then $2^{\aleph_0} < \kappa \leq 2^{\aleph_0}$. ■

Clearly we can extend ν to a measure on $\mathfrak{c}$, just let $\widetilde{\nu}(X) = \nu(X \cap \kappa)$, thus we conclude there is an atomless measure on $\mathbb{R}$. It turns out we can, with a slightly modification of the previous argument, obtain a measure on $\mathbb{R}$ extending the Lebesgue measure!

Such construction is done by partition the sets in a finer way, where we let $X_\emptyset = S$, and

$$X_{s\frown 0} \sqcup X_{s\frown 1} = X_s, \text{ with } \mu(X_{s\frown 0}) = \mu(X_{s\frown 1}) = \frac{1}{2}\mu(X_s)$$

for every binary sequence s. We then let

$$\widetilde{\lambda}(X) = \mu\Big(\bigcup\{X_s : s \in X\}\Big), \text{ where } X_s = \bigcap_{n\in\omega} X_{s\restriction_n}$$

with $\mathbb{R}$ being identified as the Cantor Space 2^ω. The work now is to verify that $\widetilde{\lambda}$ is indeed a measure, and that it agrees with the Lebesgue Measure λ on all intervals $[k/2^n, (k+1)/2^n]$, and hence on all Borel sets.

It now remains to show Ulam's main result, that every real-valued measurable cardinal is weakly inaccessible. In order to do this, we will need the following object:

Definition 2.5: Ulam Matrix

For any cardinal λ, an Ulam (λ^+, λ)-matrix is a collection $\{A_{\alpha,\eta} : \alpha < \lambda^+, \eta < \lambda\}$ of subsets of λ^+ such that:

- $A_{\alpha,\eta} \cap A_{\beta,\eta} = \emptyset$;
- $|\lambda^+ \setminus \bigcup_{\eta<\lambda} A_{\alpha,\eta}| \leq \lambda$, for each $\alpha < \lambda^+$.

In order to use it, let's guarantee that at least one Ulam (λ^+, λ)-Matrix.

Lemma 2.3

For any λ, there is an Ulam (λ^+, λ)-matrix.

Proof. Given $\xi < \lambda^+$, let f_ξ be function on λ with $\xi \subseteq \mathrm{Im}(f_\xi)$. Let then define, for $\alpha < \lambda^+$, $\eta < \lambda$,

$$\xi \in A_{\alpha,\eta} \iff f_\xi(\eta) = \alpha$$

Then, if $\alpha \neq \beta$, clearly $f_\xi(\eta)$ can't be equal both, so $A_{\alpha,\eta} \cap A_{\beta,\eta} = \emptyset$. Now, given $\alpha < \lambda^+$, let $\xi > \alpha$, so, since $\xi \subseteq \mathrm{Im}(f_\xi)$, there is some $\eta < \lambda$ such that $f_\xi(\eta) = \alpha$, i.e., $\xi \in A_{\alpha,\eta}$, therefore

$$\xi \in \bigcup_{\eta<\lambda} A_{\alpha,\eta}, \text{ for all } \xi > \alpha$$

whose complement is contained in $\alpha \leq \lambda$, as desired. ■

Let's see how Ulam Matrices breaks σ-saturation.

Theorem 2.3

Every real-valued measurable cardinal is weakly inaccessible.

Sketch: Let's do the proof for $\lambda = \omega_1$, showing there can't be a measure on it.

Intuitively, an Ulam (ω_1, ω)-matrix is a $\omega_1 \times \omega$ matrix whose entries are subsets of ω_1, positioned in such a way that:

- any two elements in a column are disjoint;
- the union of all elements in a row contains all, but countably many elements of ω_1.

In fact, this is exactly the combinatorial object we want to break σ-saturation! Since the union of all elements in a row is ω_1, except for a set of measure zero, then each row has at least one element of positive measure.

But since we have more columns than rows, then, by the Pigeonhole Principle, there is at least some column with uncountably many disjoint sets of positive measure!

Proof. We already saw that κ should be regular, then, assume by a contradiction that $\kappa = \lambda^+$. By the previous theorem there is an Ulam (λ^+, λ)-matrix.

Since the union of the sets in each row is all of λ^+, except for a set of measure 0, then each row has at least one set of positive measure. However, by the Pigeonhole Principle, since there are more rows than columns, we can find a column with uncountably many disjoint sets of positive measure, contradicting σ-saturation, so κ is a limit cardinal, implying it's weakly inaccessible. ■

As a summary of all the nice work we have seen so far, we can state the following theorem:

Theorem 2.4: (Ulam)

If there exists a measure on some set S, then either:

- there is a measure with an atom over S, and $|S|$ is $\geq$ the least inaccessible cardinal, or;
- there is an atomless measure on $\mathfrak{c}$, making it $\geq$ the least weakly inaccessible cardinal.

Therefore, the solution for the Set Theoretic Measure Problem is that: either we have a "trivial" measure on a monstrously large cardinal we can't even prove to exist in ZFC, or we have a "non-trivial" measure on $\mathfrak{c}$, making it also comically big[3].

Proof. If there is a measure on S, then either it's 2-valued, and then if $\kappa \leq |S|$ is the least cardinal with a 2-valued measure, by lemma 2.1 it's measurable, so theorem 2.1 guarantees κ is inaccessible; or it's atomless and, by theorem 2.2, there is some atomless measure on a $\lambda \leq \mathfrak{c}$. So, if κ is the least cardinal with a measure in it, by lemma 2.1 we have that κ is real-valued measurable and, by theorem 2.3, κ is weakly inaccessible, so $\mathfrak{c} \geq \lambda \geq \kappa \geq \kappa_0$ is greater or equal than the least weakly inaccessible cardinal κ_0. ■

2.2 Compacteness

[3]how beautiful are God's creations

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